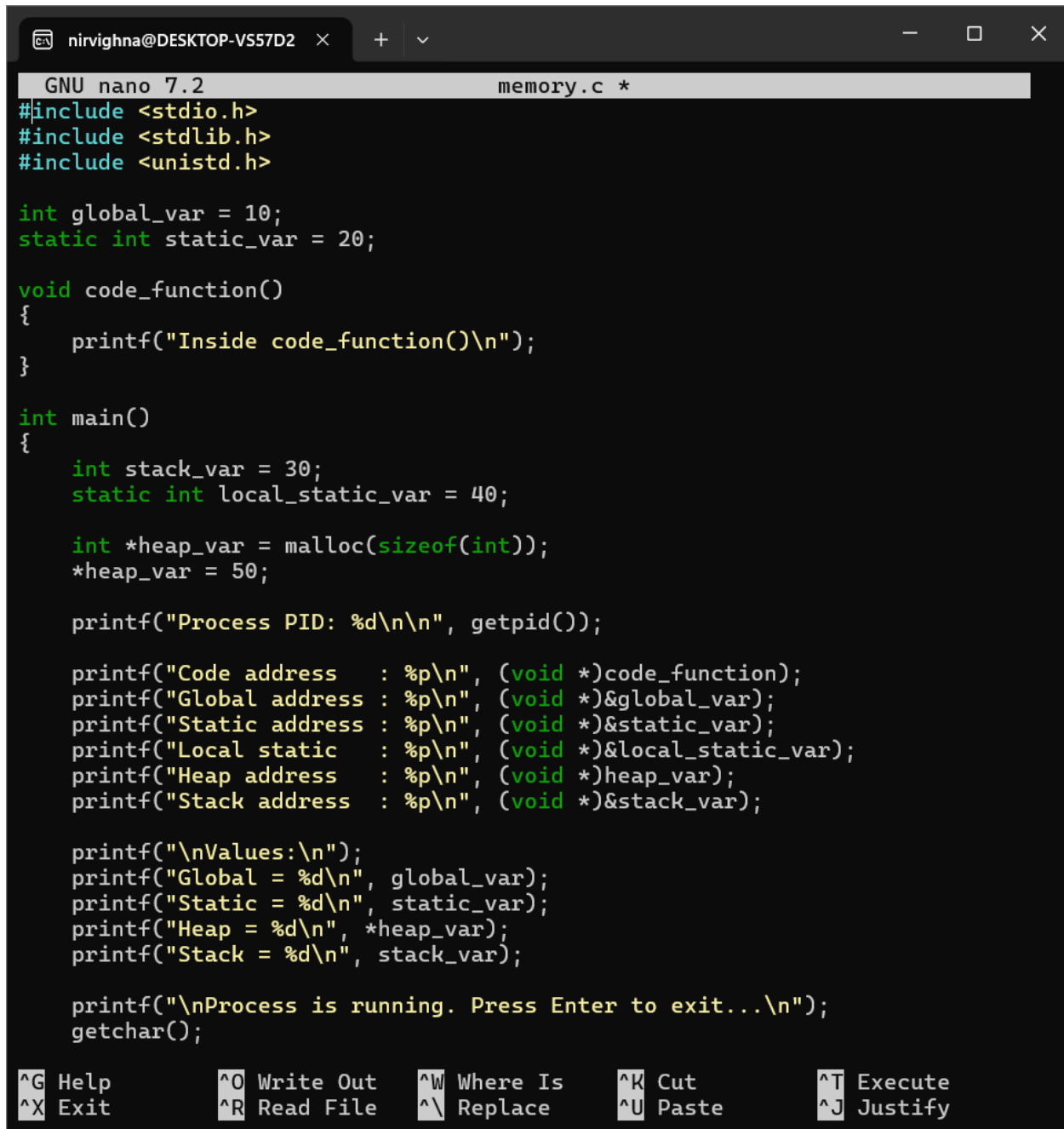


PRACTICAL-7:



```
GNU nano 7.2 memory.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("Inside code_function()\n");
}

int main()
{
    int stack_var = 30;
    static int local_static_var = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process PID: %d\n\n", getpid());

    printf("Code address   : %p\n", (void *)code_function);
    printf("Global address  : %p\n", (void *)&global_var);
    printf("Static address   : %p\n", (void *)&static_var);
    printf("Local static    : %p\n", (void *)&local_static_var);
    printf("Heap address    : %p\n", (void *)heap_var);
    printf("Stack address   : %p\n", (void *)&stack_var);

    printf("\nValues:\n");
    printf("Global = %d\n", global_var);
    printf("Static = %d\n", static_var);
    printf("Heap = %d\n", *heap_var);
    printf("Stack = %d\n", stack_var);

    printf("\nProcess is running. Press Enter to exit...\n");
    getchar();
}
```

 Help	 Write Out	 Where Is	 Cut	 Execute
 Exit	 Read File	 Replace	 Paste	 Justify

nirvighna@DESKTOP-VS57D2 X + v

```
nirvighna@DESKTOP-VS57D2I:~$ nano memory.c
nirvighna@DESKTOP-VS57D2I:~$ gcc memory.c -o memory
nirvighna@DESKTOP-VS57D2I:~$ ./memory
```

Process PID: 480

```
Code address   : 0x5c793bd3b209
Global address  : 0x5c793bd3e010
Static address  : 0x5c793bd3e014
Local static    : 0x5c793bd3e018
Heap address    : 0x5c7960c562a0
Stack address   : 0x7ffe78bd31dc
```

Values:

```
Global = 10
Static = 20
Heap = 50
Stack = 30
```

Process is running. Press Enter to exit...

|